Collective motion

Applications in studying the dynamics of tissues

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18th March, 2017

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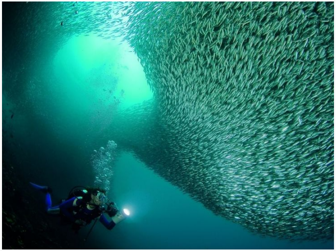
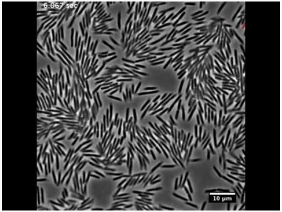
Introduction

What is collective motion?



Fish shoal

Bird flock



E. coli colony

Sheep herd

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- Such ”crowds” can be of various types, ranging from simple bacterial colonies to mobs of people in a busy street.
- The common link between such different kinds of motion is the exchange of information between the different individuals, which can be mediated in a number of ways, e.g. mechanical forces, sensory perception (vision, auditory) etc.

Why tissue dynamics?

The study of dynamics of tissues is of interest to many researchers because of a number of interesting open questions e.g.

- Morphogenesis
- Wound healing
- Cancer metastasis

Physical theories of such problems aim at quantifying these phenomena in terms of physical parameters of the corresponding biological systems.

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- The study of collective motion of self-propelled particles, draws significant parallels with already (pretty-well) understood equilibrium processes, like phase transitions
- A non-equilibrium theory of the dynamics of tissues is a more natural way to approach this problems

What I have been doing

- I have been replicating the results of some well-known models of collective motion
- Standard Vicsek model, which predicts a phase transition-like behaviour in a particular non-equilibrium system
- Collective motion with mechanical interactions

Standard Vicsek Model

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- However, the genius of the concept lies in the fact that this simple model captures the essential physical picture pretty well
- Basic principle: ”Look at neighbours at present, and take the average direction to move in, in the next time step”
- Discrete time-stepping

The model

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- Directions oriented randomly $\{\theta_i\}$

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Time stepping:

$$\mathsf{x}_i(t + \Delta t) = \mathsf{x}_i(t) + \mathsf{v}_i \Delta t$$

$$\theta_i(t+1) = \langle \theta(t) \rangle_r + \zeta$$

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- $\langle \theta(t) \rangle_r$ is given by the angle $\arctan(\langle \sin\theta \rangle_r / \langle \cos\theta \rangle_r)$

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- ρ, the density of particles in the periodic square domain of length L, so that we have $\rho = \frac{N}{L^2}$
- v_0, the distance traveled by each self-propelled particle (SPP) in a single time step

Order parameter

We define a kinetic order parameter v_a as follows

$$v_a = \frac{1}{N v_0} \left| \sum_{i=1}^{N} \mathbf{v}_i \right|$$

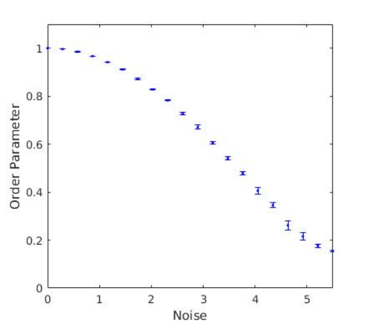
In a disordered state $v_a \approx 0$, while in an ordered state, we should expect $v_a \approx 1$
This order parameter is analogous to the magnetization M of a ferromagnetic sample, where M is large at low temperatures while it drops to 0 at higher temperatures

Results

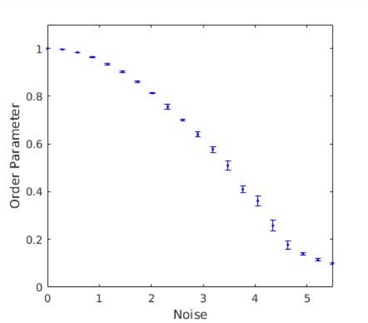
The results of simulations with this model, by varying the noise parameter η, for different number of particles N are discussed next. The parameters used are as follows

- Drift speed $v_0 = 0.03$
- Interaction radius $r = 1$
- Time step $\Delta t = 1$
- Density $\rho = 4$

Results ($\rho = 4$)

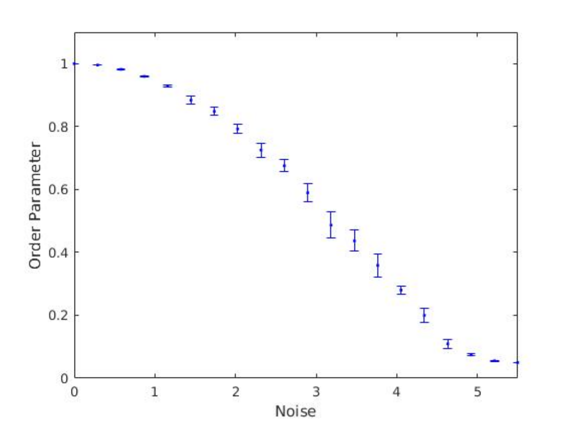


$N = 40$



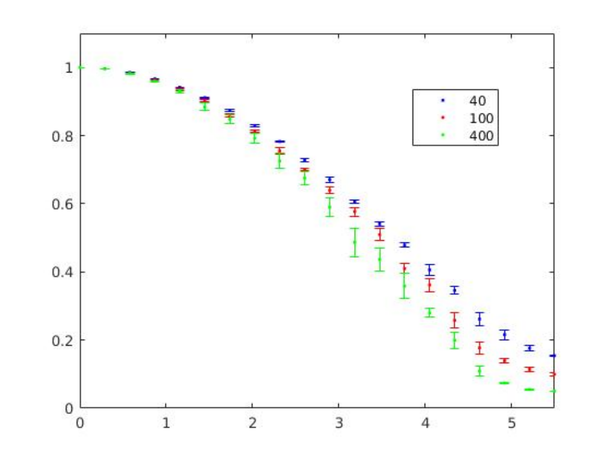
$N = 100$

Results ($\rho = 4$)



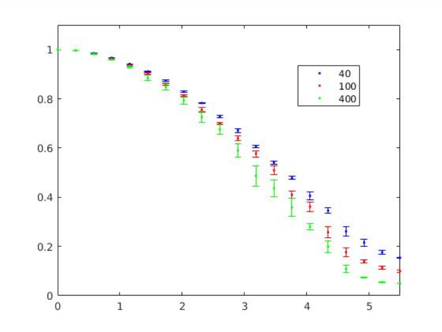
$N = 400$

Results ($\rho = 4$)

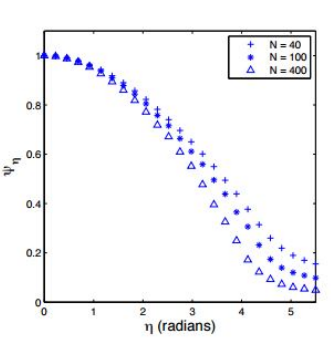


Combined simulation results

Comparison of results with paper



Simulation Results



From Lee, Couzin et al (2011)

Noise: Extrinsic vs Intrinsic

- The way that the stochastic noise term is included in Vicsek model is called "intrinsic" noise, distinguishing it from "extrinsic" noise
- A continuous phase transition in the order parameter was seen when intrinsic noise was used
- In presence of extrinsic noise, on the other hand, some studies have found a discontinuous (first-order) phase transition in the order parameter

Mechanically intermediated interactions

Mechanical interactions

- The mode of interaction described in some of the models such as the one described by Szabo et al (2006) involve explicit forces, unlike the Vicsek model
- The interaction force is a simple spring force, with a finite cut-off radius.

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- The interaction force is attractive at longer separations, repulsive at small distances
- The particles have an excluded volume, unlike the point particles in the standard Vicsek model
- A phase transition can be observed here similar to that seen with the Vicsek model

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Time stepping:

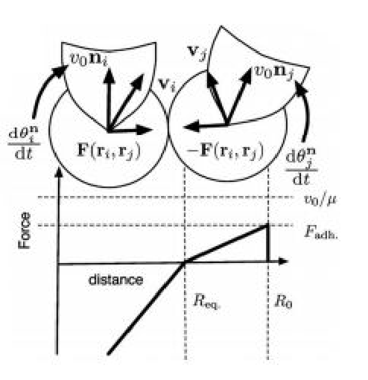
$$\frac{d\mathbf{r}_i}{dt} = v_0 \mathbf{n}_i(t) + \mu \sum_{j=1, j\neq i}^{N} \mathbf{F}(\mathbf{r}_i, \mathbf{r}_j)$$

$$\frac{d\theta_i^n(t)}{dt} = \frac{1}{\tau} \arcsin\left[\left(\mathbf{n}_i \times \frac{\mathbf{v}_i(t)}{||\mathbf{v}_i(t)||}\right) \cdot \hat{\mathbf{e}}_z\right] + \xi$$

Here, ξ is a delta-correlated gaussian ”white” noise term, τ is the relaxation time of the alignment, and μ is the mobility of a given particle

The model

The force of interaction can be graphically portrayed as follows



Interaction force

Simulation (closed domain)

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- Link to the simulation video

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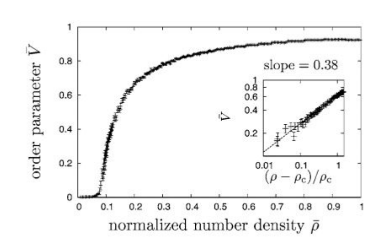
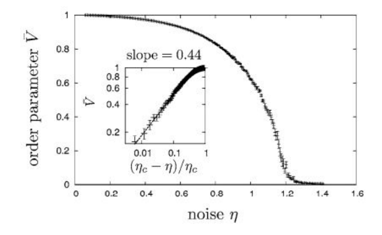
- This would let us check if a phase transition behaviour is seen in the order parameter, on changing the noise strength, or on changing the density
- It has been found *[Szabo et al (2006)]* that these phase transitions have critical exponents which are nearly equal to the values obtained from the Vicsek model

Results from paper

Results from *Phase transition in the collective migration of tissue cells: Experiment and model (Szabo et al.)* are presented here.



Noise strength fixed at $\eta = 0.6$



Normalized density fixed at $\bar{\rho} = 0.6$

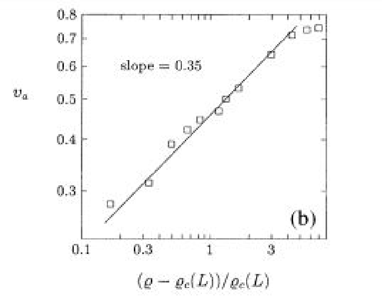
Results from paper

The main conclusions

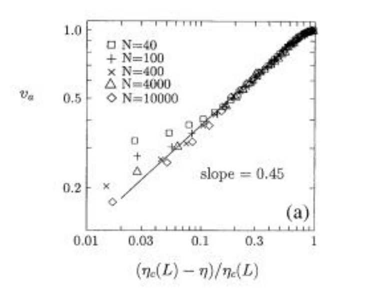
- The order parameter is clearly seen to show a second order (continuous) phase transition, as the noise strength or the density of particles is varied
- As discussed before, this phase transition is found to have critical constants that are nearly equal to the ones obtained by *Vicsek et al (1995)*

Phase Transition in Vicsek model

The following are the critical constants evaluated by analyzing the Vicsek model [*Source: Vicsek et al (1995)*]



Order parameter scaling with density



Order parameter scaling with noise

Conclusion

The road ahead

- Use these models of collective motion in modeling epithelial cells, to study the deformation, stresses etc. that are generated in tissues that are undergoing change, e.g. in wound-healing process
- Jamming, a transition to a state of low mobility, could provide a way to stop the metastasis of cancerous cells *(Quanta Magazine: Jammed Cells Expose the Physics of Cancer)*

References

1. Vicsek, T., Czirók, A., Ben-Jacob, E., Cohen, I., & Shochet, O. (1995). Novel type of phase transition in a system of self-driven particles. Physical review letters, 75(6), 1226.
2. Szabo, B., Szöllösi, G. J., Gönci, B., Jurányi, Z., Selmeczi, D., & Vicsek, T. (2006). Phase transition in the collective migration of tissue cells: experiment and model. Physical Review E, 74(6), 061908.
3. Vicsek, T., & Zafeiris, A. (2012). Collective motion. Physics Reports, 517(3), 71-140.
4. Quanta Magazine (16th August, 2016 Issue) Jammed Cells Expose the Physics of Cancer